Magnetic Noise Enabled Biocompass

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Also thank Prof. Haiguang Liu at CSRC for sharing the knowledge of the MagR, and acknowledge for the discussions with many other professors.

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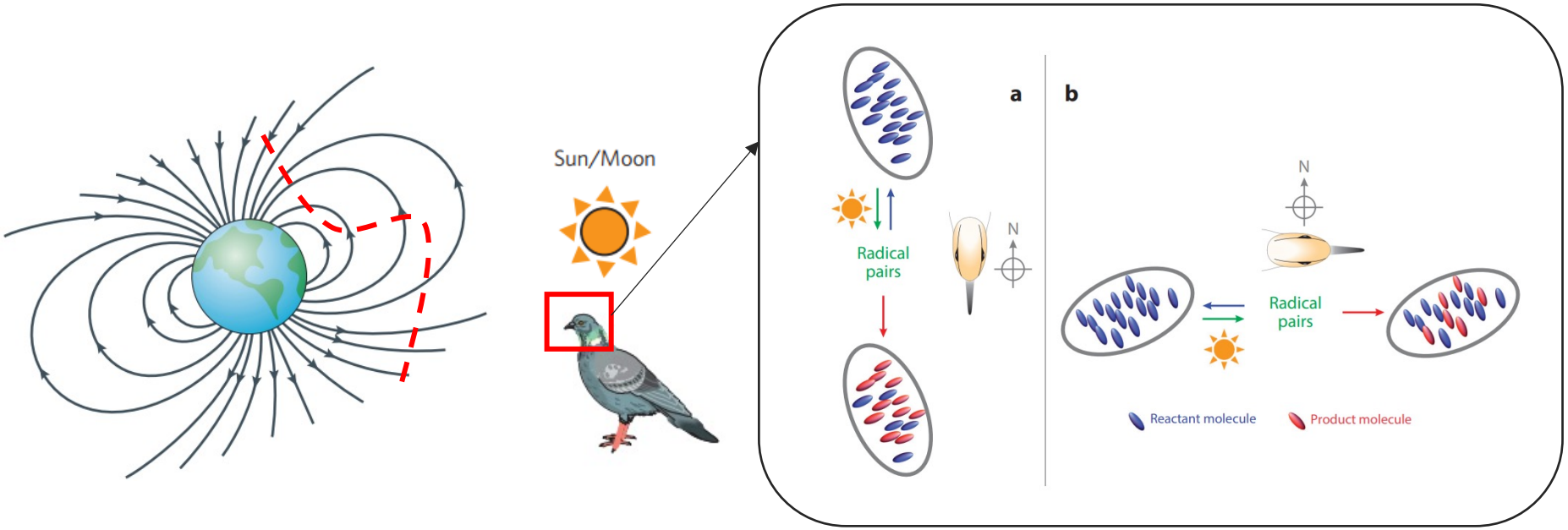
Biocompass: the radical pair model



radical pairs in geomagnetic field (in birds' eyes)
control the **efficiency** of the reactant-product conversion

K. Schulten and P. G. Wolynes, J. Chem. Phys. **68**, 3292 (1978).
P. J. Hore and H. Mouritsen, Annu. Rev. Biophys. **45**, 299 (2016).
S. Qin et.al. Nat. Mater. **15**, 217 (2016).

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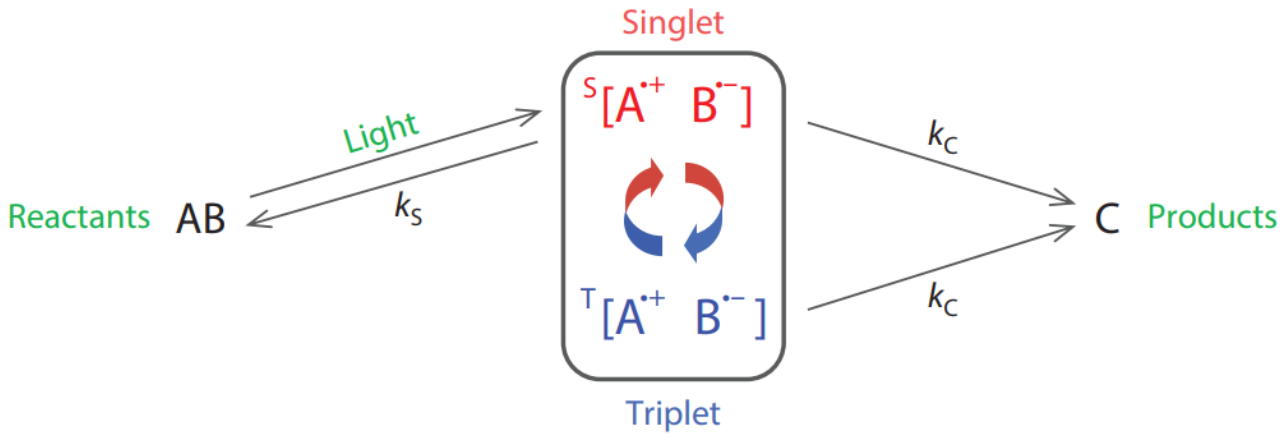
Biocompass: spin singlet and triplets



Spin Singlet

$$|S\rangle = (|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)/\sqrt{2}$$

Spin Triplets

$$|T_+\rangle = |\uparrow\uparrow\rangle$$

$$|T_0\rangle = (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle)/\sqrt{2}$$

$$|T_-\rangle = |\downarrow\downarrow\rangle$$

The radical pair spin singlet and triplets dynamics controlled the efficiency of the reactants-products conversion

↓

geomagnetic sensitive

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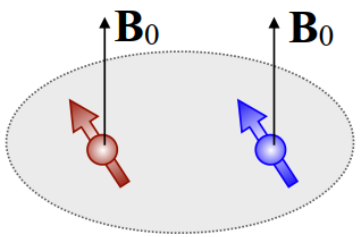
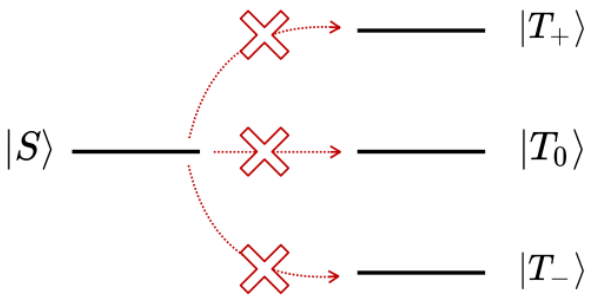
Biocompass: inhomogeneous field

homogeneous field

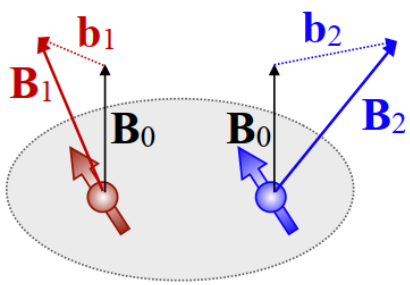


$$H = -\gamma \mathbf{B}_0 \cdot (\mathbf{S}_1 + \mathbf{S}_2)$$

total spin conservation



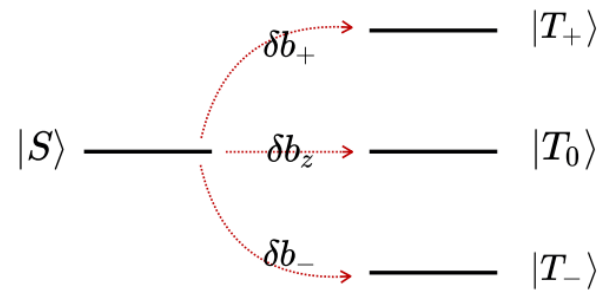
inhomogeneous field



$$H = -\gamma \mathbf{B}_0 \cdot (\mathbf{S}_1 + \mathbf{S}_2) - \gamma (\mathbf{b}_1 \cdot \mathbf{S}_1 + \mathbf{b}_2 \cdot \mathbf{S}_2)$$

singlet-triplet conversion

$$\delta \mathbf{b} = \mathbf{b}_1 - \mathbf{b}_2$$



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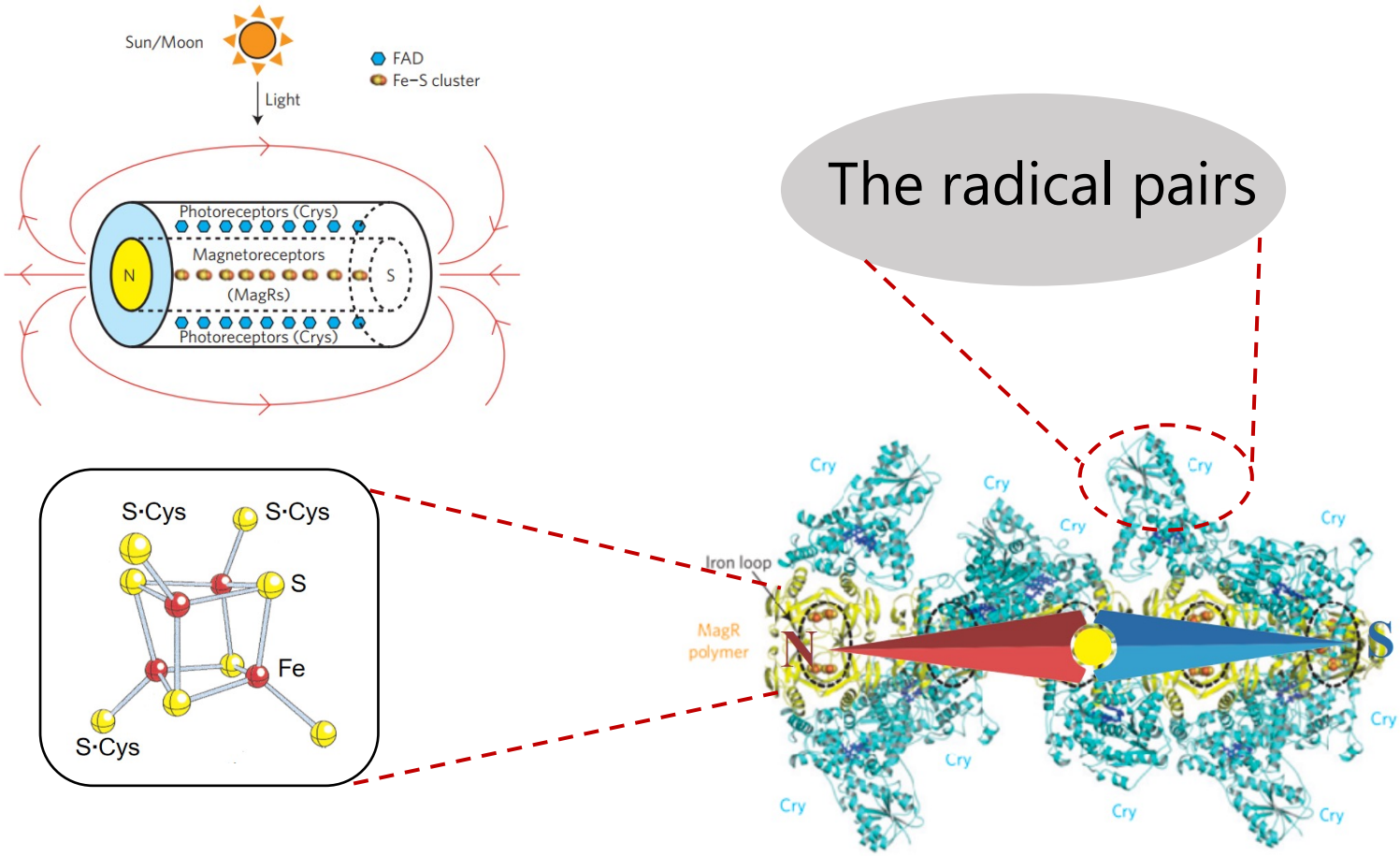
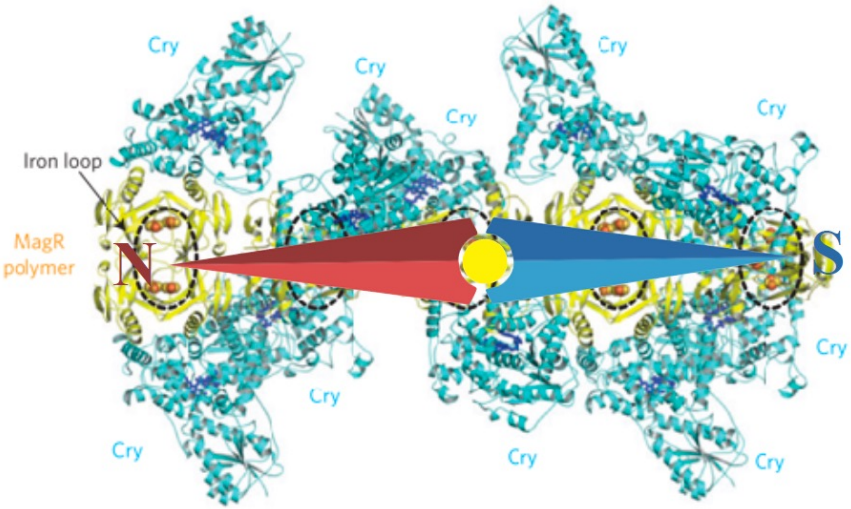
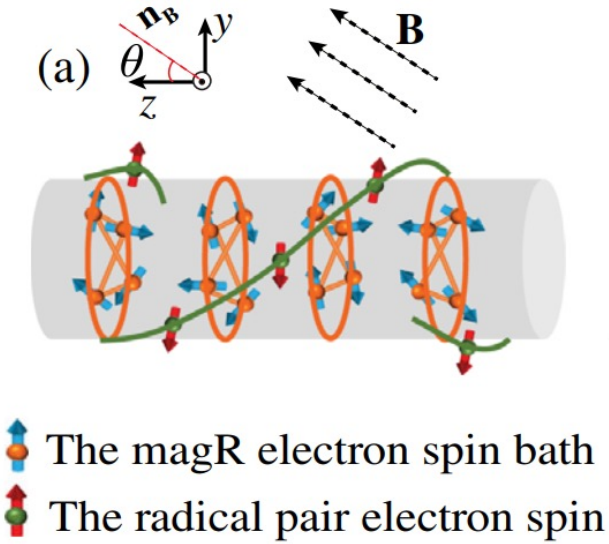
Biocompass: MagR

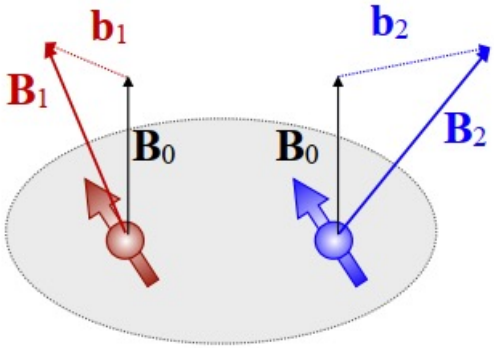


[1] S. Qin et.al. Nat. Mater. **15**, 217 (2016).
[2] H. Beinert, R. H. Holm, and E. Münck, Science **277**, 653 (1997)

Biocompass: MagR







$b_1 \neq 0, b_2 \neq 0$

$b_1 \neq b_2$

"Intrinsic magnetic polarity of the magnetosensor" [1]

[1] S. Qin et.al. Nat. Mater. **15**, 217 (2016).

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Biocompass: debates

Physical limits to magnetogenetics

Markus Meister*

Division of Biology and Biological Engineering, California Institute of Technology, Pasadena, United States

Abstract This is an analysis of how magnetic fields affect biological molecules and cells. It was prompted by a series of prominent reports regarding magnetism in biological systems. The first claims to have identified a protein complex that acts like a compass needle to guide magnetic orientation in animals (*Qin et al., 2016*). Two other articles report magnetic control of membrane conductance by attaching ferritin to an ion channel protein and then tugging the ferritin or heating it with a magnetic field (*Stanley et al., 2015*; *Wheeler et al., 2016*). Here I argue that these claims conflict with basic laws of physics. The discrepancies are large: from 5 to 10 log units. If the reported phenomena do in fact occur, they must have causes entirely different from the ones proposed by the authors. The paramagnetic nature of protein complexes is found to seriously limit their utility for engineering magnetically sensitive cells.

DOI: 10.7554/eLife.17210.001

- room temperature magnetization
- room temperature orientation

permanent magnetization requires

- ~10^6 closely packed iron atoms, or
- low temperature < 14 Kelvin

$$\frac{\text{magnetization energy in geomagnetic field}}{\text{thermal energy}} = \frac{mB_{\text{Earth}}}{kT} = 2 \times 10^{-5}$$

M. Meister, eLife **5**, e17210 (2016).

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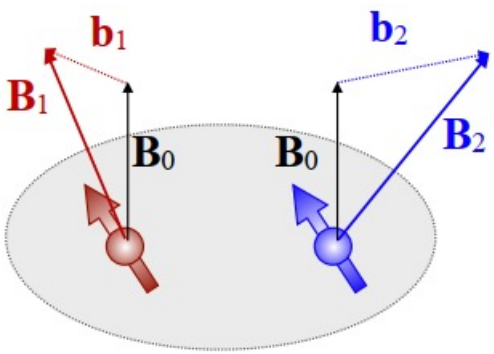
Biocompass: debates



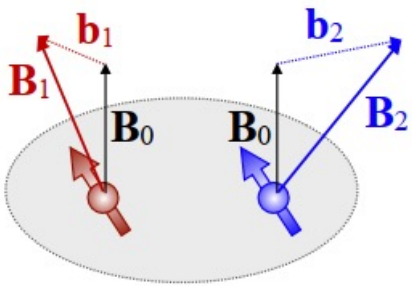
$$\langle \mathbf{b}_1 \rangle = \langle \mathbf{b}_2 \rangle = 0$$

$$\langle \delta \mathbf{b} \rangle = 0$$

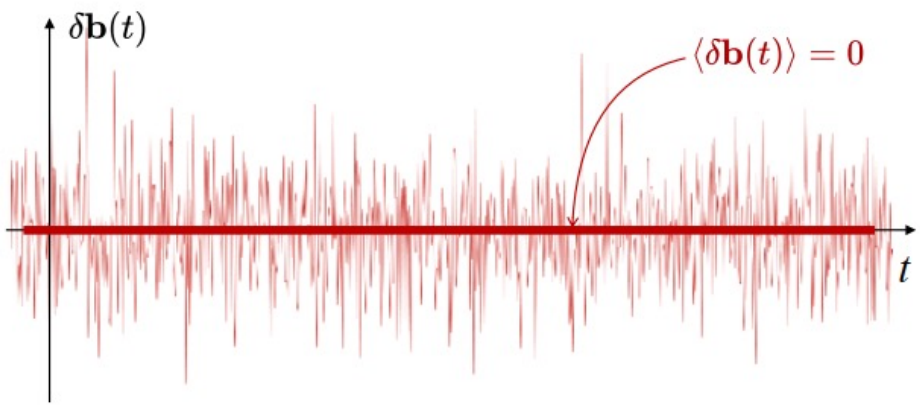
$\langle \cdot \rangle$ thermal average

There is no magnetic polarity!

Magnetic fluctuations



$$\delta\mathbf{b} = \mathbf{b}_1 - \mathbf{b}_2$$



Can **higher order correlations** of $\delta \boldsymbol{b}(\boldsymbol{t})$ affect the radical pair spin dynamics?

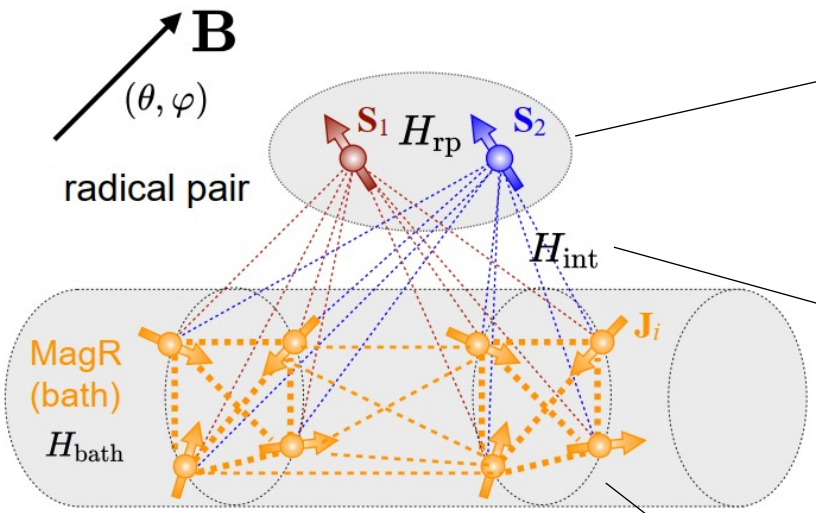
1st order (mean value) $\langle\delta\mathbf{b}(t)\rangle = 0$

2nd order (fluctuation) $\langle\delta\mathbf{b}(t_1)\delta\mathbf{b}(t_2)\rangle = ?$

Variance $\langle\delta\mathbf{b}^2\rangle > 0$

.....

Magnetic fluctuations



$$H = H_{\rm rp} + H_{\rm bath} + H_{\rm int}$$

$$H_{\rm rp} = \sum_k \omega_k \left|\phi_k\right\rangle \left\langle\phi_k\right| ,$$

$$\text{for } \left|\phi_k\right\rangle \in \{\left|S\right\rangle, \left|T_0\right\rangle, \left|T_+\right\rangle, \left|T_-\right\rangle\}$$

$$H_{\rm int} = \sum_{k,i} \mathbf{S}_k \cdot \boxed{\mathbb{A}_{ki} \cdot \mathbf{J}_i} \equiv \sum_{k=1,2} \gamma_e \mathbf{S}_k \cdot \mathbf{b}_k$$

$$\mathbf{b}_1 = \sum_{i=1}^{N} \mathbb{A}_{1i} \cdot \mathbf{J}_i / \gamma_e \qquad \mathbf{b}_2 = \sum_{i=1}^{N} \mathbb{A}_{2i} \cdot \mathbf{J}_i / \gamma_e$$

$$H_{\rm bath} = \sum_{i=1}^{N} \gamma_i \mathbf{B} \cdot \mathbf{J}_i + \sum_{i>j=1}^{N} \mathbf{J}_i \cdot \mathbb{D}_{ij} \cdot \mathbf{J}_j$$

inhomogeneous field

$$\delta\mathbf{b} = \mathbf{b}_1 - \mathbf{b}_2 = \sum_{i=1}^{N} (\mathbb{A}_{1i} - \mathbb{A}_{2i}) \cdot \mathbf{J}_i / \gamma_e$$

$$[\delta\mathbf{b}, H_{\rm bath}] \neq 0$$

$$\delta\mathbf{b}(t) = e^{iH_{\rm bath}t} \delta\mathbf{b} e^{-iH_{\rm bath}t}$$

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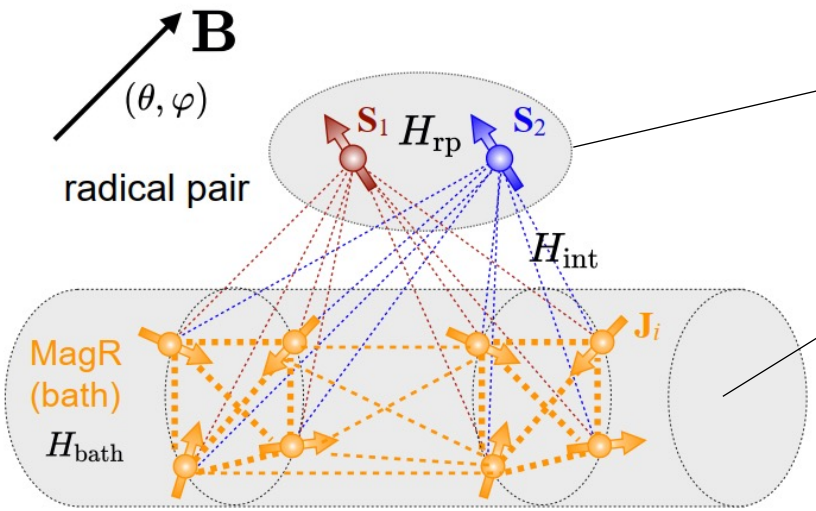
Magnetic fluctuations



$$\rho_{rp}(t=0) = |S\rangle\langle S|$$

$$\rho_{bath}(t=0) = \bigotimes_{i=1}^{N} \frac{\mathbb{I}_i}{\mathrm{Tr}[\mathbb{I}_i]}$$

$\mathbb{I}_i$ identity matrix

$$H = H_{rp} + H_{bath} + H_{int}$$

1st order (mean value)

$$\langle \delta\mathbf{b}(t)\rangle = \mathrm{Tr}[\rho_{bath}(0)\delta\mathbf{b}(t)] \equiv 0$$

unpolarized spin bath

2nd order (fluctuation)

$$C(\tau) = \langle \delta\mathbf{b}(t)\delta\mathbf{b}(t+\tau)\rangle$$
$$= \mathrm{Tr}[\delta\mathbf{b}(t)\delta\mathbf{b}(t+\tau)\,\rho_{bath}(0)]$$

power spectrum ↓ F.T.

$$\mathcal{S}_\alpha(\omega) = \frac{1}{2^N}\sum_{m,n} |\langle\psi_m|\delta b_\alpha|\psi_n\rangle|^2 \delta(\omega - \epsilon_{mn})$$

(Noise spectrum)

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Magnetic fluctuation

$$\mathcal{S}_\alpha(\omega) = \frac{1}{2^N}\sum_{m,n} |\langle\psi_m|\delta b_\alpha|\psi_n\rangle|^2 \delta(\omega - \epsilon_{mn})$$

Transitions in the spin bath

$$H_{\text{bath}}|\psi_m\rangle = \epsilon_m|\psi_m\rangle$$

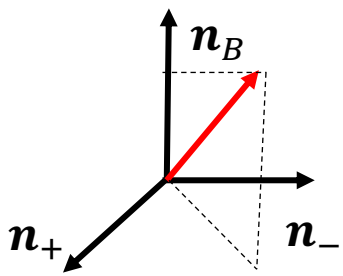
Structures of the MagR spin

B-coordinates projection on inhomogeneous field

$$\delta b_\alpha = \delta\boldsymbol{b}\cdot\boldsymbol{n}_\alpha,\ \alpha = 0, \pm$$

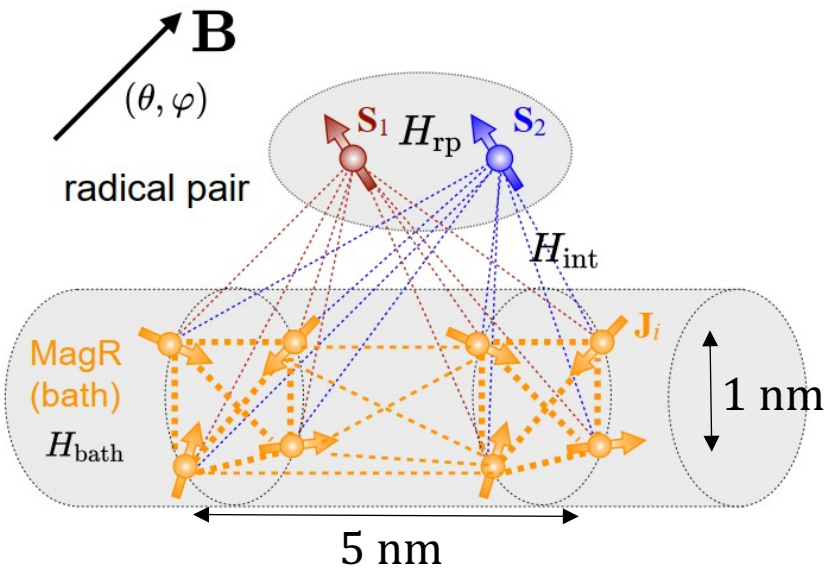
$$\mathbf{n}_\pm = (\mp\,\mathbf{e}_x - i\mathbf{e}_y)/\sqrt{2}$$

$$\boldsymbol{n}_0 = \boldsymbol{n}_B$$



Microscopic structures ⟷ Magnetic fluctuations

Model



$$H_{rp} = \sum_k \omega_k |\phi_k\rangle\langle\phi_k| \quad \text{for } |\phi_k\rangle \in \{|S\rangle, |T_{0,\pm}\rangle\}$$

$$H_{int} = \sum_{k,i} \mathbf{S}_k \cdot \mathbb{A}_{ki} \cdot \mathbf{J}_i \equiv \sum_{k=1,2} \gamma_e \mathbf{S}_k \cdot \mathbf{b}_k$$

$$H_{bath} = \sum_{i=1}^{N} \gamma_i \mathbf{B} \cdot \mathbf{J}_i + \sum_{i>j=1}^{N} \mathbf{J}_i \cdot \mathbb{D}_{ij} \cdot \mathbf{J}_j$$

anisotropicity

$$\mathbb{A}_{ki} = \frac{\mu_0 \gamma_e^2 \hbar (1 - 3\hat{\mathbf{r}}_{ki}\hat{\mathbf{r}}_{ki})}{r_{ki}^3}$$

$$\mathbb{D}_{ij} = \frac{\mu_0 \gamma_e^2 \hbar (1 - 3\hat{\mathbf{r}}_{ij}\hat{\mathbf{r}}_{ij})}{r_{ij}^3}$$

Strength

$B \sim 0.5\,\text{Gauss}$ Geomagnetic field

$$\frac{\mu_0 \gamma_e \hbar}{4\pi r^3} = \begin{cases} 18.6\,\text{Gauss} & r \sim 1\text{nm} \\ 0.15\,\text{Gauss} & r \sim 5\text{nm} \end{cases}$$

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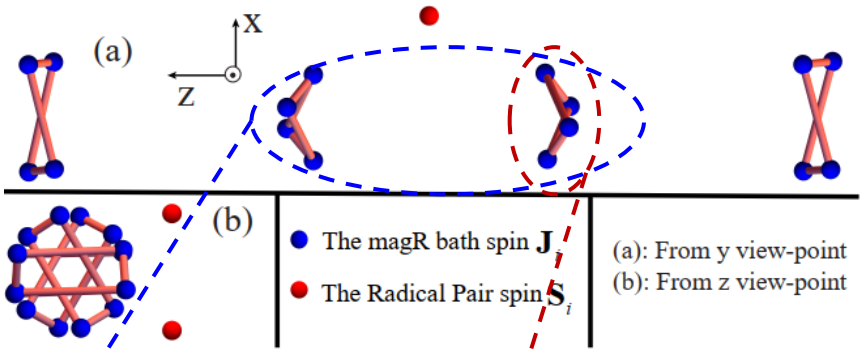
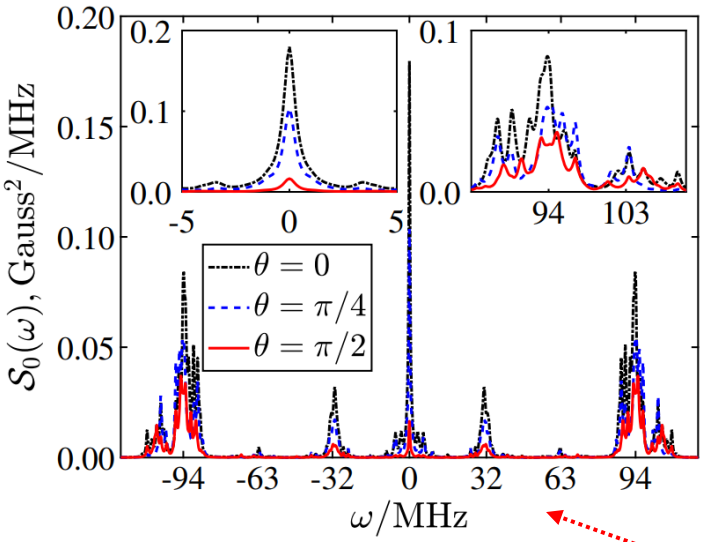
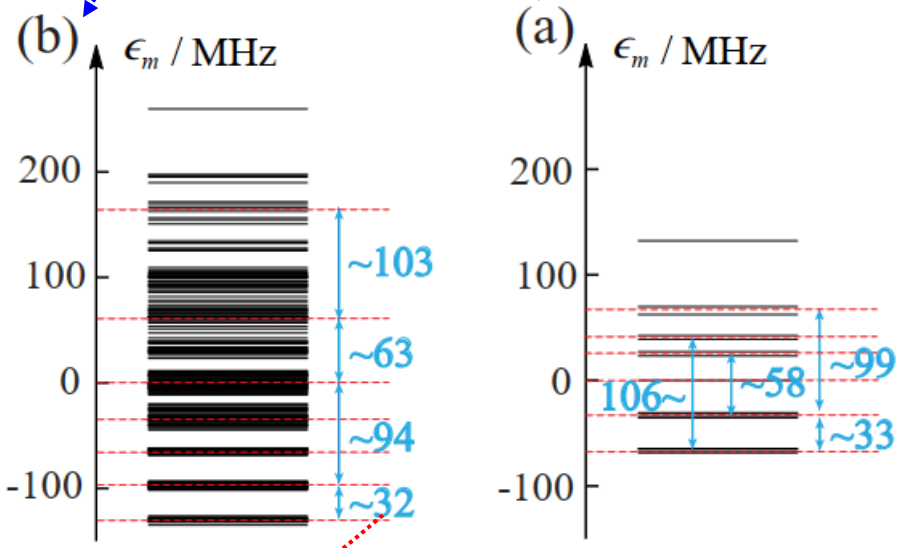
Model

$$\mathcal{S}_\alpha(\omega) = \frac{1}{2^N} \sum_{m,n} \left| \langle \psi_m | \delta b_\alpha | \psi_n \rangle \right|^2 \delta(\omega - \epsilon_{mn})$$



Noise spectrum

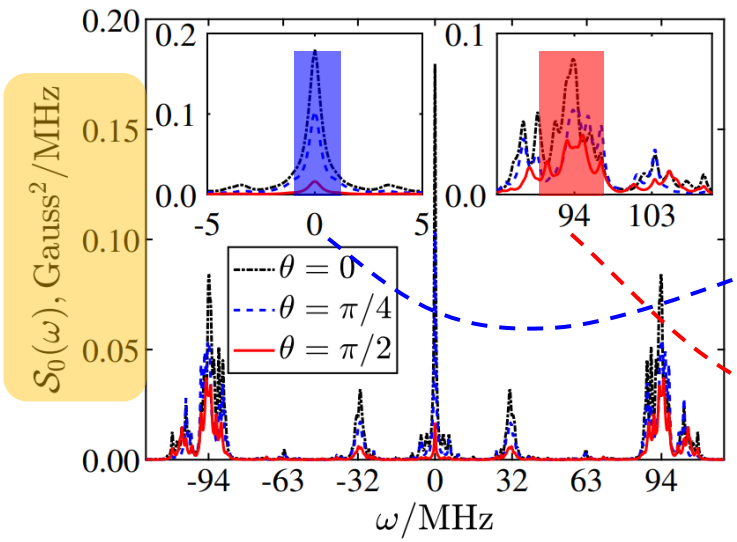


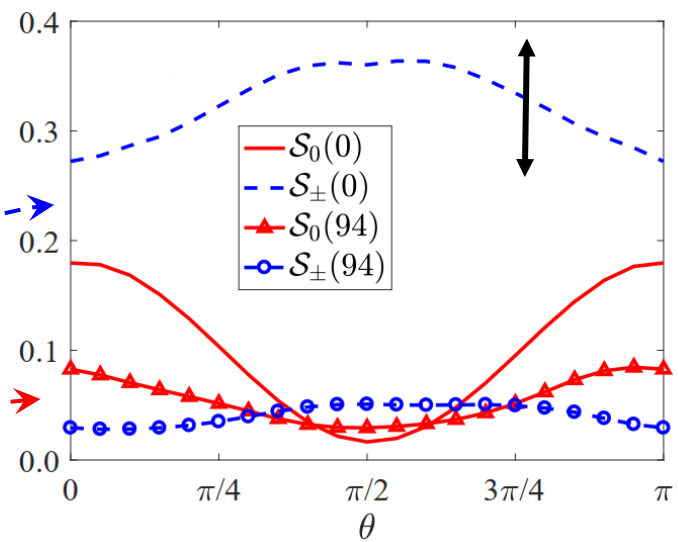


Model

$$\mathcal{S}_\alpha(\omega) = \frac{1}{2^N}\sum_{m,n} |\langle \psi_m | \delta b_\alpha | \psi_n \rangle|^2 \delta(\omega - \epsilon_{mn})$$

Noise spectrum





Magnetic noise strength ⟶ Comparable to geomagnetic field

Magnetic Noise anisotropicity ⟶ Sensitive to geomagnetic field

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Spin singlet dynamics

$$\rho(0) = \rho_{\mathrm{rp}}(0) \otimes \rho_{\mathrm{bath}}(0)$$

Evolution

$$\rho(t) = e^{-iHt}\rho(0)\,e^{iHt}$$

Singlet fidelity

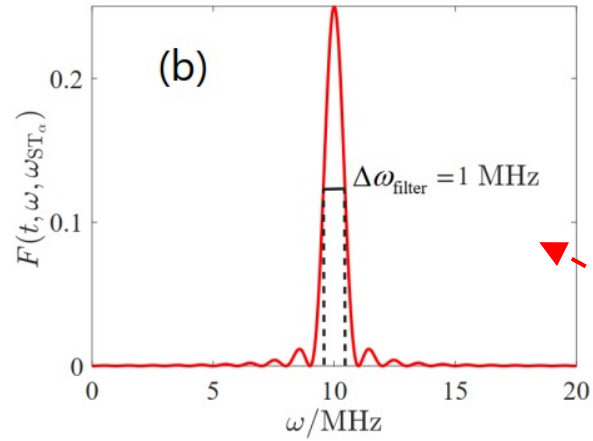
$$P_S(t) = \mathrm{Tr}\left[|S\rangle\langle S|\rho(t)\right]$$

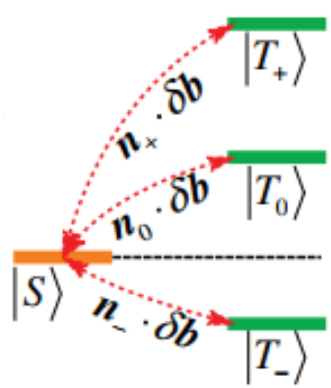
Singlet productivity

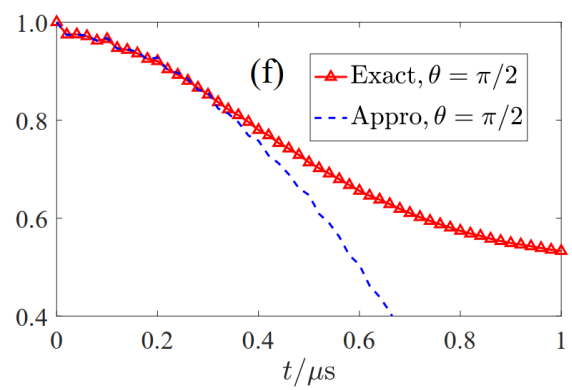
$$\Phi_S(\tau) = \frac{1}{\tau}\int_0^{\tau} P_S(t)\,dt$$

Short-time ($\leq 1\mu s$)

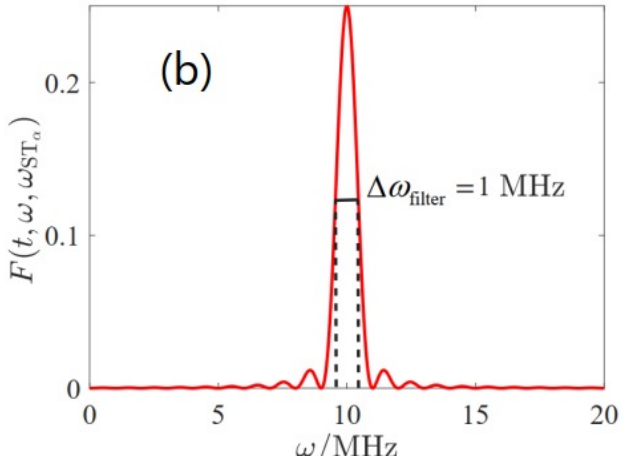
$$P_S(t) \approx 1 - \sum_{\alpha=0,\pm} \gamma_e^2 \int_{-\infty}^{\infty} S_\alpha(\omega)\, F\left(t,\omega,\omega_{ST_\alpha}\right) d\omega$$



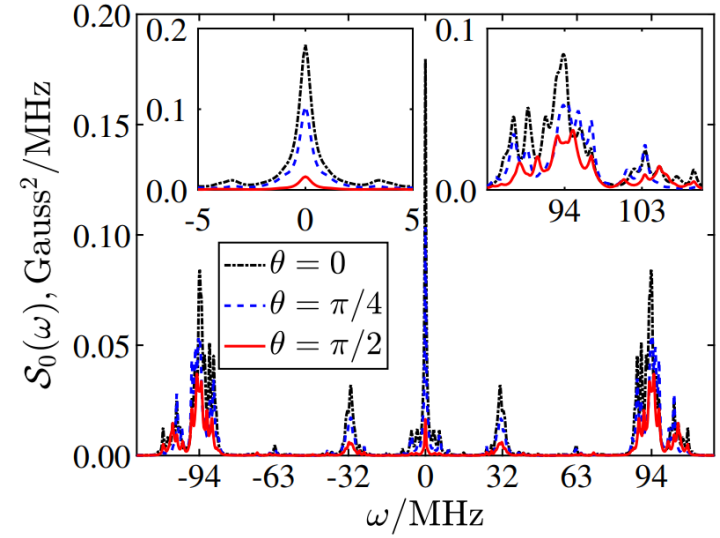


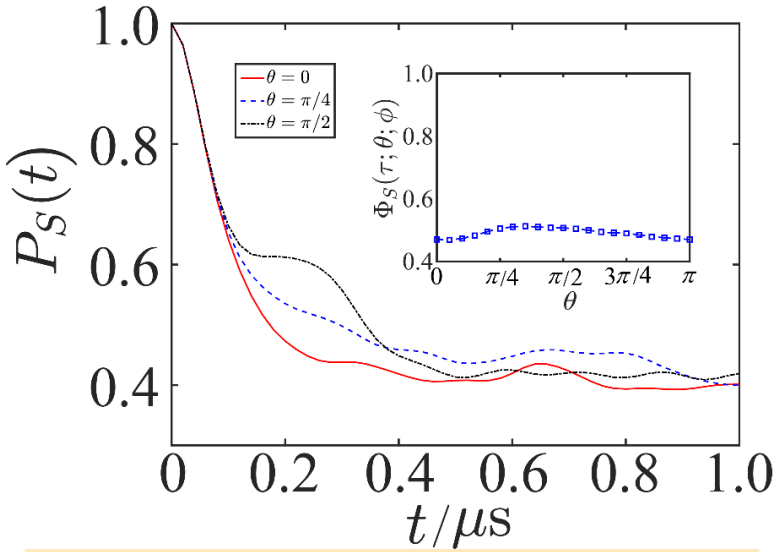


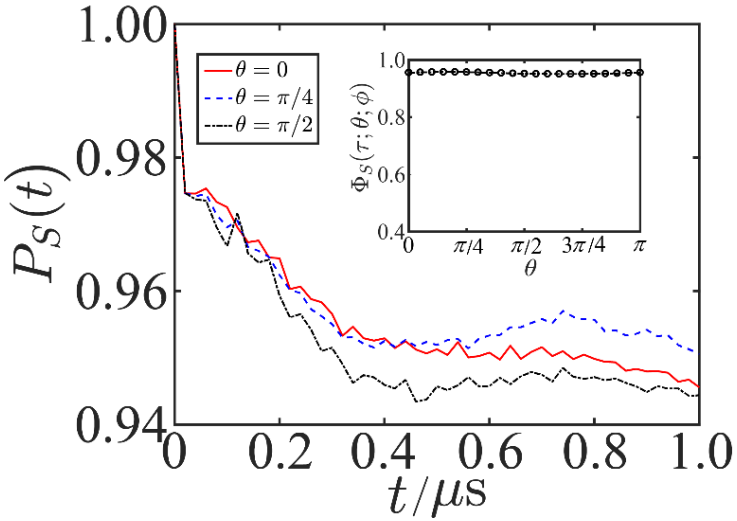
Resonance



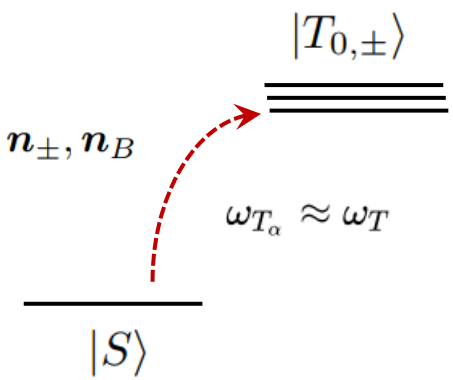
Overlap







Non-degeneracy

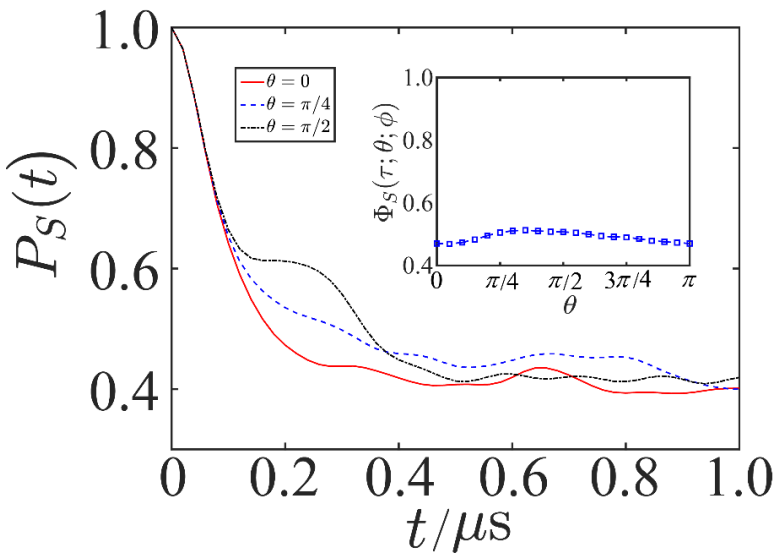


$$P_S(t) = 1 - \sum_\alpha P_{S\to T_\alpha}(t)$$

$$P_S(t) \approx 1 - \gamma_e^2 \int_{-\infty}^{\infty} \left(\sum_\alpha \mathcal{S}_\alpha(\omega) \right) F(t, \omega, \omega_{ST})\, d\omega$$

$$\sum_\alpha \mathcal{S}_\alpha(\omega) = \frac{1}{2^N} \sum_{m,n} |\langle \psi_m | \delta \mathbf{b} | \psi_n \rangle|^2 \delta(\omega - \epsilon_{mn})$$

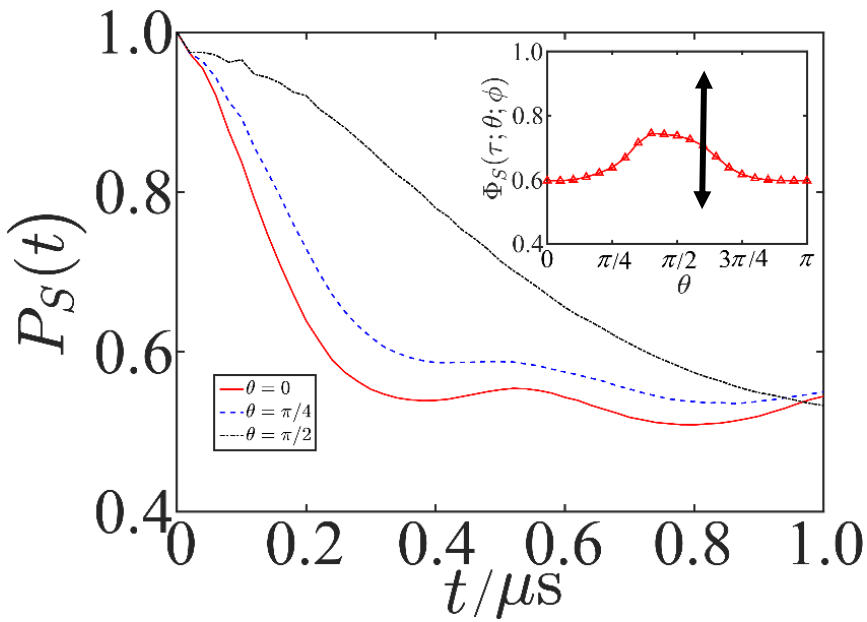
Less sensitive to geomagnetic field direction

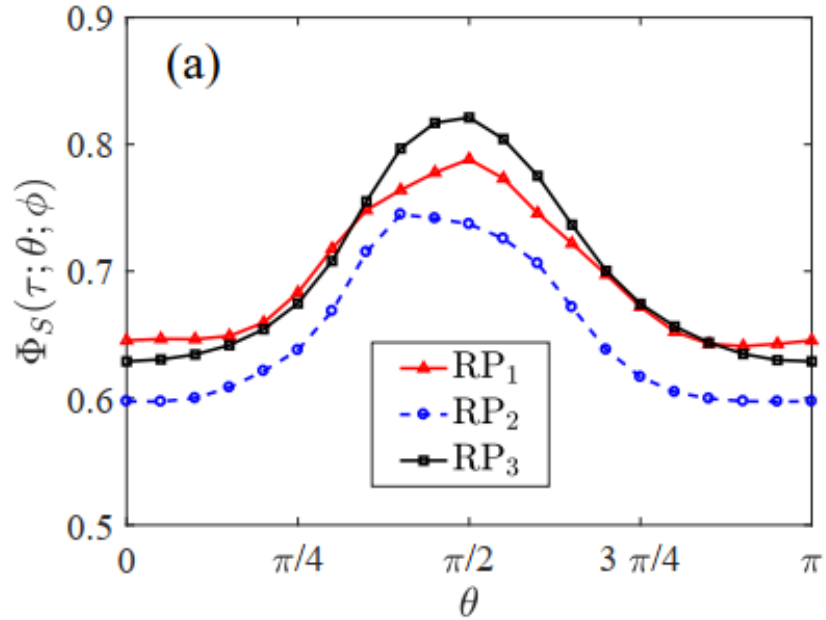


Condition (ii): Non-degeneracy

Triplet states should be non-degenerate, so that the summation-rule is broken.

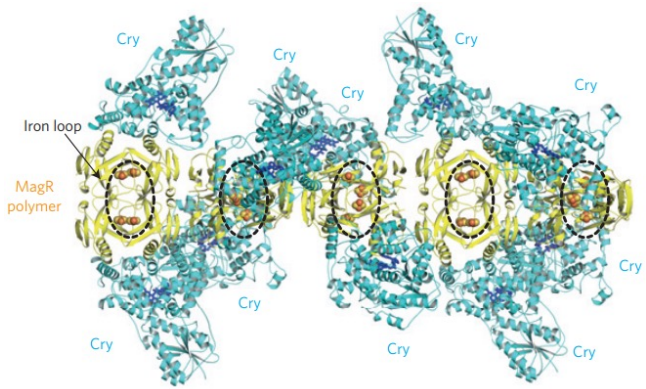
Examples



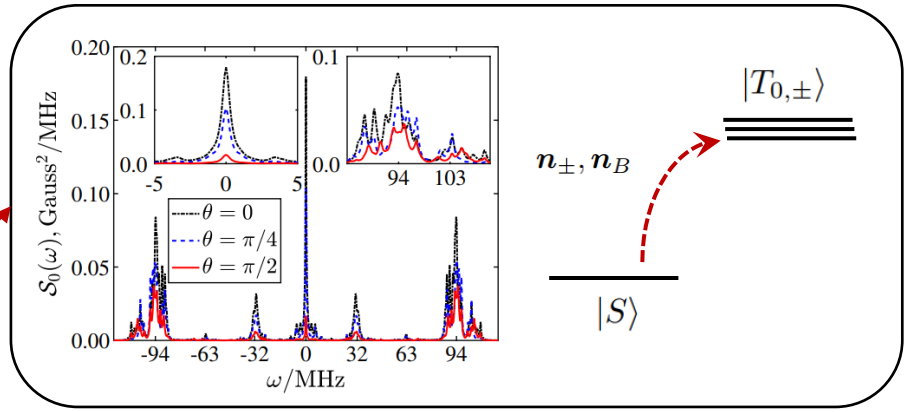


Examples to functionalize the biocompass after fulfilling above conditions!

Summary



The electron spin bath (e.g. magR) is able to functionalize the biocompass even at room temperature.



Experiment criteria

Prerequisite

- Resonance
- Non-degeneracy
- Coherence → Ongoing story

Da-Wu Xiao, Wen-Hui Hu, Yunfeng Cai and Nan Zhao Phys. Rev. Lett. **124**, 128101 (2020).

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Thank You

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